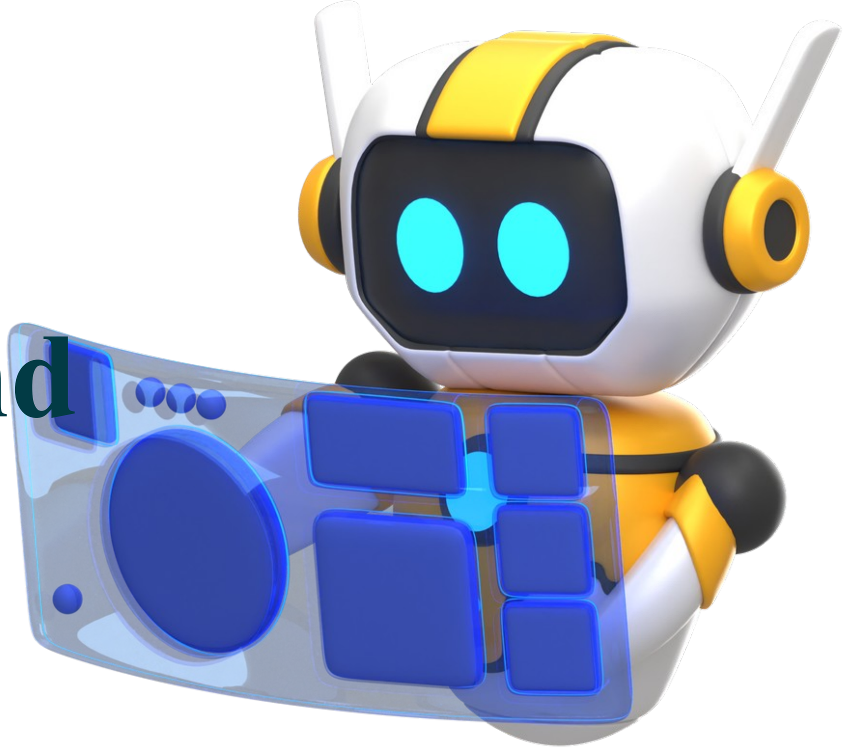
MINISTRY OF MINING INDUSTRY AND GEOLOGY

“INSTITUTE OF GEOLOGY AND GEOPHYSICS NAMED AFTER H.M. ABDULLAEV” SUE

“PRESIDENT AI AWARD” competition

PETROLOG AI — AI-powered mineralogical and petrographic analysis platform



YULDOSHEV ZOKIR DONIYOR OGLI

PROJECT TEAM

YULDOSHEV ZOKIR
DONIYOR OGLI

Junior researcher.

Project lead

Yoshuzoqov Shohboz Gaybulloyevich

Thin-section data analyst

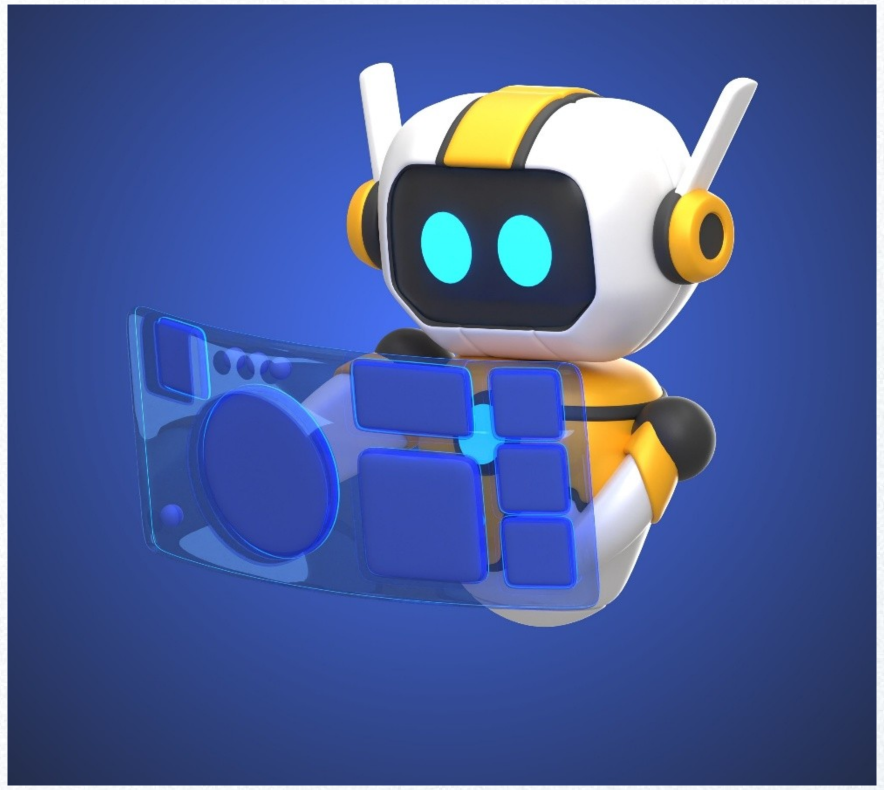
Keldibekova Zilola Asror qizi

Responsible for office-based and laboratory work within the project

STATE OF THE ART

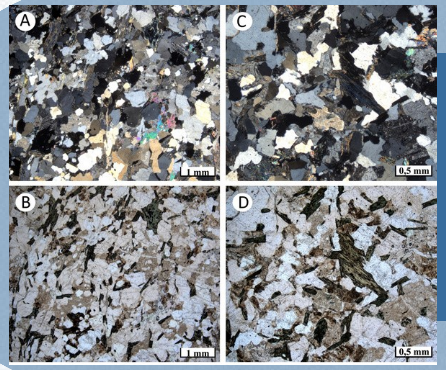
- In recent years, significant progress has been made worldwide in applying artificial intelligence to geological analysis. In particular, Computer Vision and Deep Learning methods have gained considerable importance in mineralogical and petrographic workflows, both in research and in industry. This work is well represented in the international literature. In countries leading in geoscience — the USA, Canada, Australia, China and Japan — results achieved with CNN (Convolutional Neural Network) models for microscopic image analysis are recognised by specialists in the field.
- Petrological research currently relies on tools such as PetroExplorer and Petroledge, and on resources including Alex Strekeisen, Pax-it, Petrog, FastGAPP and Petra. We use them for building diagrams and for processing thin- and polished-section photographs. However, all of these tools operate on static reference data and none of them integrates thin section, polished section and ICP data. Modern research demands complex, multi-source analysis and results aligned with international standards.
- Most existing developments remain at the research stage: they are not fully integrated into industrial practice and lack a usable interface. Petrolog AI addresses this gap by combining mineral identification from microscopic images, integration of geochemical analysis, and automated report generation within a single platform. The project is the first large-scale attempt to introduce such an AI approach into Uzbek geology.

PROJECT RELEVANCE



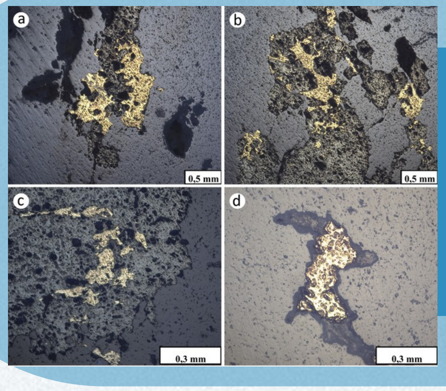
The project is the first in Uzbekistan to integrate artificial intelligence with the geosciences, meeting a pressing national need in science and industry through the creation of a digital laboratory system.

Petrolog AI is therefore not only an innovative project but one of strategic importance, capable of answering both present and future needs.



Thin section

Primary source for identifying mineral composition, texture and microstructure



Polished section

Used to study ore minerals and their mutual relationships

ICP-MS analysis

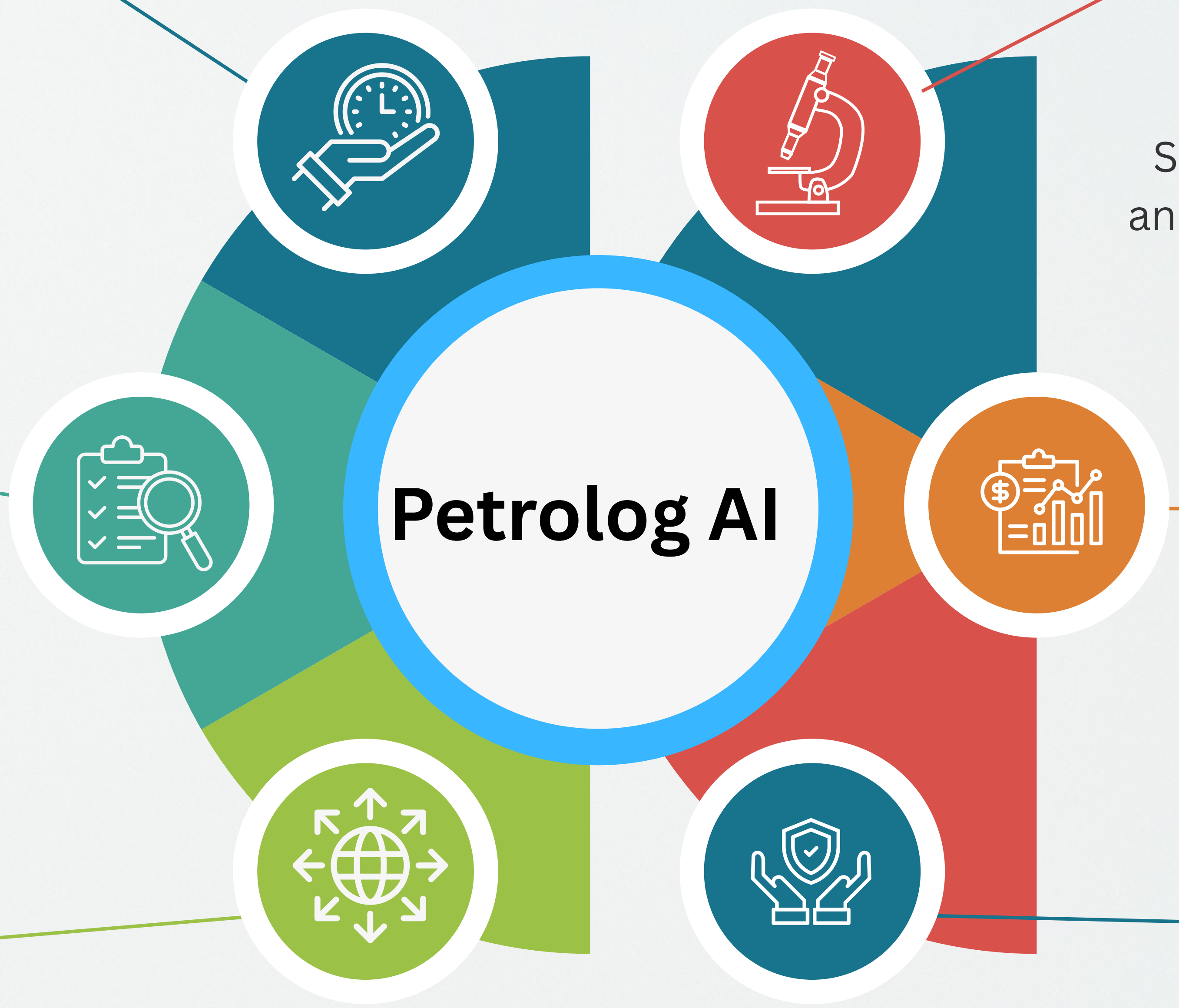
Geochemical classification and petrogenetic interpretation by elemental composition

Exploration work and contracts

12 geological exploration projects and 27 commercial contracts completed.

Practical value of the project

Petrolog AI substantially increases practical efficiency by automating research and production workflows in geological analysis. The platform enables automatic AI-based analysis of thin- and polished-section images captured under the microscope, which delivers the following key benefits



1

Time and resources saved

Analyses that traditionally take hours or days are completed in minutes with Petrolog AI

2

Higher consistency

An AI-based model reduces human subjectivity and ensures consistency across analytical results.

3

Methods grounded in international practice

The system draws on internationally grounded reference data to produce petrographic conclusions

4

Valuable for education and research

Students and researchers can carry out complex analyses quickly and accurately, raising the quality of their coursework and scientific output

5

Adaptable to industry

Rapid mineral identification accelerates production decisions in mining, mineral resources, construction materials and drilling operations.

6

Modern approach

Petrolog AI supports the digitalisation of the geological sector in Uzbekistan and contributes to modernising the national analytical infrastructure by linking science and technology

Project goals and objectives

The main goal of Petrolog AI is to determine the material composition of rocks, describe them and interpret them geologically, by analysing thin- and polished-section images captured under the microscope together with chemical analysis results using artificial intelligence

Intended goals of the project

01 A digital laboratory system is established

02 Mineralogical and petrographic analysis is automated

03 Time and resources saved when working with geological data

04 A modern tool for research and industrial needs is created

Priority objectives

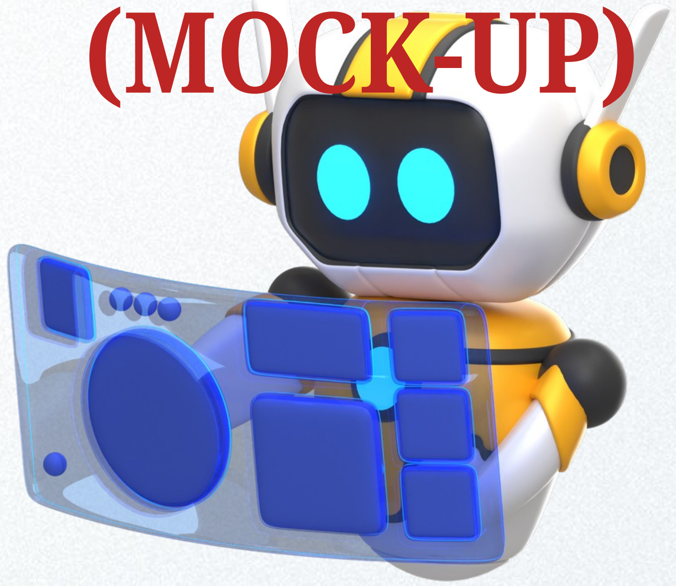
05 Build an AI model that analyses thin- and polished-section images together with supporting data

06 Develop algorithms that identify minerals from images

07 Build a simple and convenient user interface

08 Produce scientific and practical recommendations from analytical results

PETROLOG AI INTERFACE (MOCK-UP)



AI ANALYSIS MODES

- Thin section
- Polished sec.
- ICP data
- Complex

Upload thin-section image

Note: Format TIFF, PNG, JPG, max 10 MB, must include a scale marker*

Upload polished-section image

Note: Format TIFF, PNG, JPG, max 10 MB, must include a scale marker*

Upload ICP-MS results

Note: Format CSV, XLSX, max 10 MB*

Upload 360° video of polished section

Note: Format MP4, MOV, max 20 MB, must be a full 360° rotation*

AI analysis

Search

Home

Analysis

Results

Contact

Uploading

Thin section
Done

Polished sec.
Wait

ICP file
Error

Section video
Done

AI analysis modes

- Thin section
- Polished sec.
- ICP data
- Complex

Video guide

Analysis progress 75%

Sample metadata

Parameter	Value
Sample code	ZZ-1
Rock type	Granodiorite
Microscope	Nikon ECLIPSE LV 100
Objective	20× (thin), 40× (polished)

Scientific description generated from AI analysis

The sample consists of quartz (48.6%), plagioclase (23.4%) and biotite (16.1%), with amphibole and magnetite present as accessory ore minerals.
Optical features correspond to an intermediate intrusive rock — granodiorite. Sericitisation is observed in plagioclase and hydrobiotite alteration in biotite. ICP results indicate SiO_2 of 62.8%, consistent with the intermediate range. An Fe/Mg ratio of 1.86 suggests a moderate degree of metamorphic alteration.
Overall, the integrated AI analysis assigns the sample to a granodiorite intrusive complex.

AI analysis results

Mineral	Vol. %	Conf.	Pleochr.	Optical feature	Note
Quartz	48.6	0.97	none	full extinction	relatively corroded
Plagioclase	23.4	0.91	weak	zonal structure	sericitised
Biotite	16.1	0.93	strong	green–brown	altered
Amphibole	7.8	0.84	moderate	yellow-green	elongate crystals
Ore minerals	4.1	0.79	—	metallic lustre	fine disseminated

Summary

- AI classification: Granodiorite
- International reference: consistent with the IUGS (Streckeisen, 1973) diagram
- AI confidence: illustrative value
- Expert verification: yes
- Mock-up — live demo linked in README

Export ISO/ASTM

Upload doc.

ISO/ASTM LABORATORY REPORT FORMAT

Hujjat kodi: GGI 05/2025

Standart asos: ISO/IEC 17025:2017, ASTM D2798-20, ISO 14577-1:2015

Hisobot turi: Kompleks petrografik va kimyoviy tahlil

1. Umumiy ma'lumotlar

Parametr	Ma'lumot
Laboratoriya	Petrograf AI — AI asosida geologik tahlil markazi
Hisobot raqami	05/2025 loyiha
AI modeli	**Petrolog AI** v0.4 (PyTorch, multimodal fusion)
Ekspert tasdiqlovchi	____________________

2. Namuna ma'lumotlari

Parametr	Qiymat
Namuna kodi	ZZ - 1
Tog' jinsi turi	**Granodiorit**
Mikroskop modeli	Nikon ECLIPSE lv 100
Ob'ektiv kattaligi	20× (shlif), 40× (anshlif)

3. AI tahlil natijalari

Mineral	Ulushi (%)	Aniqlik (conf.)	Pleoxroizm	Optik belgi	Izoh
Kvars	48.6	0.97	Yo'q	To'liq aniqlik	Nisbatan yedirilgan
Plagioklaz	23.4	0.91	Kam	Zonal tuzilish	Seritsitlashgan
Biotit	16.1	0.93	Kuchli	Qizg'ish-jigarrang - sarg'ish	O'zgargan
Amfibol	7.8	0.84	O'rtacha	Yashil-ko'k pleoxroizm	Uzun kristallarga ega
Ma'danli minerallar	4.1	0.79	—	Metallik yaltirash	Mayda tarqoq donachalardan iborat

4. AI tahlilidan generatsiya qilingan ilmiy tavsif

Namuna kvarts (48.6%), plagioklaz (23.4%) va biotit (16.1%)dan tashkil topgan bo'lib, amfibol va magnetit ma'danli minerallari qo'shimcha komponent sifatida ishtirok etadi.

Optik belgilar oraliq kislotali intruziv jins — **granodiorit** tarkibiga mos keladi. Plagioklazda serikitleanish, biotitda gidrotit almashinuvi kuzatiladi. ICP natijalari SiO_2 62.8% bo'lishini ko'rsatadi, bu oraliq kislotali diapazonga mosdir. Fe/Mg nisbati 1.86 — metamorfik o'zgarishning o'rtacha bosqichini bildiradi. Umuman, AI-integratsiya natijasi bo'yicha namuna **granodiorit intruziv kompleksiga** mansub.

Xulosa

- AI klassifikatsiyasi: **Granodiorit**
- Xalqaro moslik: IUGS (Streckeisen, 1973) diagrammasi bilan mos
- AI ishonchliligi: 88.8%
- Ekspert tasdiqlash: **Ha**
- Model versiyasi: v0.4 (FusionNet)

Lavozim	F.I.Sh.	Imzo
AI tizimi operatori	J. Xudoyberdiyev	______
Ekspert geolog	G. Xolmurodov	______
Laboratoriya rahbari	N. Usmonova	______

INTEGRATED SCIENTIFIC AND ANALYTICAL TEXT OUTPUT

Tahlil natijasi:

AI yordamida tahlil qilingan SHLIF-2025-021 namunasi optik va kimyoviy ma'lumotlarning kompleks integratsiyasiga asoslanadi. Shlif tasvirida kvarts (taxminan 49%) asosiy komponent sifatida qayd etilgan bo'lib, u past relyefli, aniq interferensiya ranglariga ega, pleoxroizm belgilari kuzatilmagan. Plagioklaz (23%) seritsitlanish belgilari bilan aniqlanadi; u zonal tuzilishga ega bo'lib, qisman Ca–Na almashinuvi ko'rinadi. Biotit (16%) kuchli pleoxroizm (qizg'ish-jigarrang → sarg'ish) bilan ajralib turadi va qisman gidrotitga aylangan. Amfibol (8%) yashil-ko'k rangli pleoxroizmga ega, uzun kristalli shaklda joylashgan. Ruda minerallar (4%) pirit va magnetit ko'rinishida reflektiv yuzalarda aniqlangan.

ICP-MS tahlilida SiO_2 miqdori 62.8% bo'lib, bu tog' jinsni nordon tarkibli ekanini ko'rsatadi. Al_2O_3 (16.3%), Fe_2O_3 (5.2%) va MgO (2.8%) nisbati jinsning biotit–amfibol bilan boyigan tarkibini ko'rsatadi. Fe/Mg nisbati 1.86 bo'lib, bu metamorfik o'zgarishning o'rtacha bosqichini bildiradi. K/Na nisbati 0.83 bo'lishi K-feldspat va plagioklaz orasidagi muvozanatni ko'rsatadi.

AI modeli tomonidan hosil qilingan integratsiyalangan fazoviy xususiyatlar (segmentatsiya maskalari, rang tahlili, kimyoviy vektorlar) birgalikda interpretatsiya qilinib, namunaning tarkibi **granodiorit intruziv** jinslariga mosligi aniqlangan. Shlifda kuzatilgan biotit–amfibol birikmalari va ICP natijasidagi Fe/Mg nisbati metamorfik qayta kristallannish belgilari bilan mos keladi.

Namuna **IUGS Streckeisen (1973)** diagrammasida granodiorit sohasi ichiga tushadi. AI modelining o'rtacha ishonchlilik darajasi 0.888 bo'lib, ekspert bahosi bilan moslik koeffitsienti 0.82 ni tashkil etdi.

Xulosa: SHLIF-2025-021 — kvarts–plagioklaz–biotit–amfibol tarkibli, nordon intruziv tog' jinsi. Petrografik va kimyoviy tahlillar kompleks xolatda ushbu namuna granodiorit ekanligini ko'rsatadi.

MARKET SIZE

Bottom-up estimate: annual cost of petrographic labour → share a software product can capture

55,000

Petrographic analyses performed annually in Uzbekistan (estimate)

$3.96M

Annual cost of the expert labour spent on that work

$475,000

TAM — Uzbekistan (software market, 12% capture)

$1.19M

TAM — including Central Asia

Cross-check by number of organisations

- 40 potential customer organisations: mining companies, Uzbekgeologiya units, research institutes, universities
- Average annual contract value ≈ $11,900 — consistent with our price list

Strategic conclusion

The Uzbek market alone is small — roughly half a million dollars a year. Regional expansion and adjacent applications (core logging, ore quality control) are therefore not an optional upside but a strategic necessity.

Note: the figure of 55,000 is an extrapolation. Official statistics will be requested from the Ministry.

BUSINESS MODEL AND PRICING

Annual licence plus on-premise deployment. Most cost is one-off;each marginal customer carries a high margin.

Small laboratory

$4,800

per year

fewer than 3,000 samples per year

Medium laboratory

$14,000

per year

3,000–10,000 samples per year

Large / on-premise

$42,000

per year

Model adaptation, support, data never leaves the site

Unit economics — why a customer pays

$72

Manual: 12 hours × $6

$10

With platform: 0.5 h + fee

86%

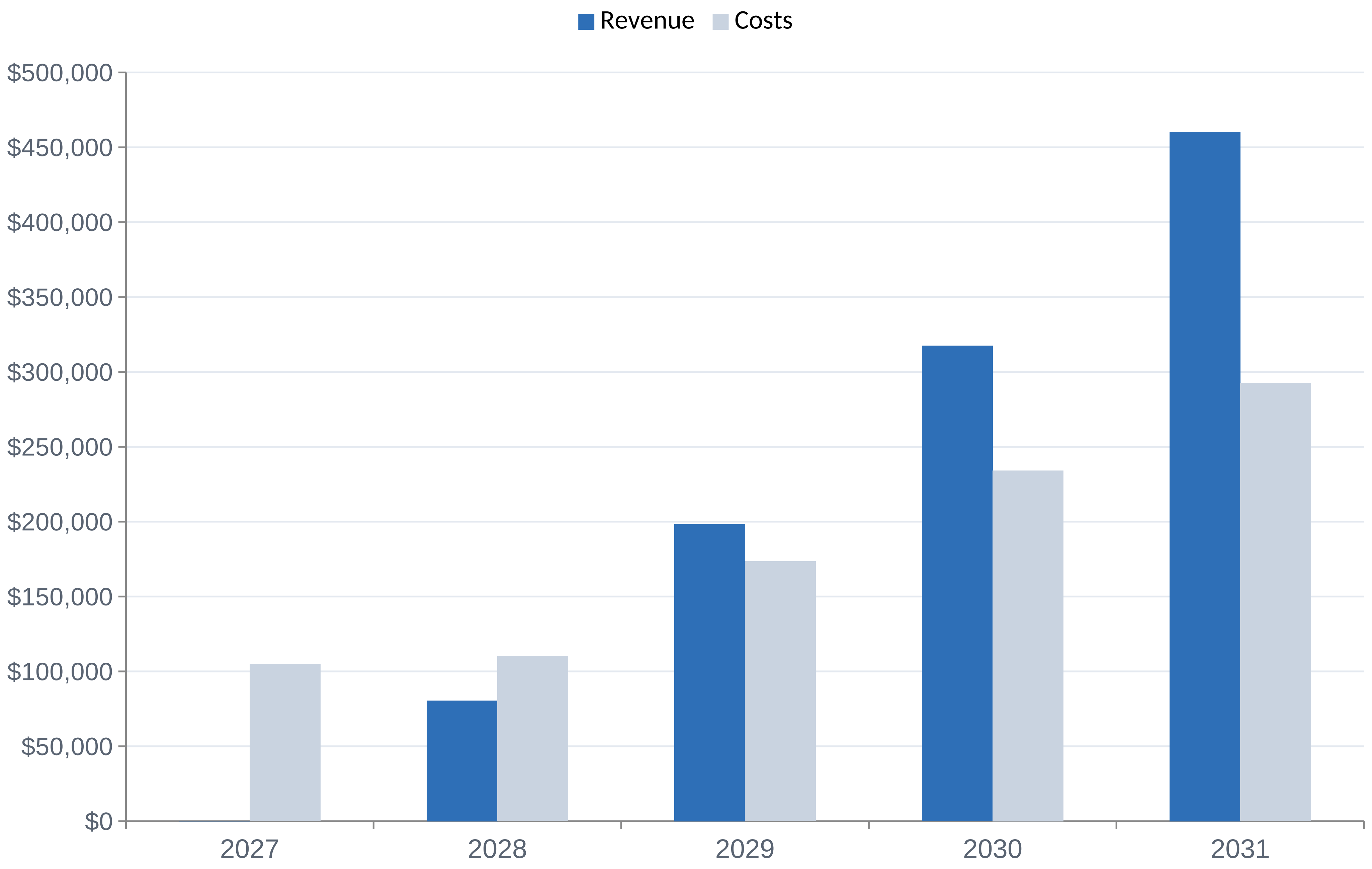
Customer saving

96%

Our gross margin

FINANCIAL PROJECTION

Conservative scenario. Every figure is computed with formulas in the attached financial model.



$13,600 NPV (5 years, 20% discount)

24.5% IRR

2029 Break-even

$135,000 Peak funding need

The $100,000 award covers most of the peak funding requirement; the remainder comes from partnerships and service revenue.

12-MONTH ROADMAP

Every stage has a measurable outcome. Accuracy targets will be set after the pilot dataset, not before.

Months 1–3	**Dataset core** Digitise 20+ research reports, write the annotation protocol	≥1,500 annotated images
Months 4–6	**First model** Train segmentation and classification models	F1 ≥ 0.75 on major rock-forming minerals
Months 7–9	**Closed beta** Testing inside the institute laboratory workflow	≥300 samples processed through the system
Months 10–12	**Pilot customer** Deployment at an industrial partner	≥1 signed pilot; ≥60% time saving

TEAM — WHAT WE HAVE AND WHAT IS MISSING

We hold the domain expertise and the data. The award is needed to close the engineering gap.

Existing core

Yuldoshev Zokir Doniyor ogli

Project lead, junior researcher

Yoshuzoqov Shohboz Gaybulloyevich

Thin-section data analyst

Keldibekova Zilola Asror qizi

Office-based and laboratory work

Roles to be hired

ML engineer

Model architecture, training, evaluation

Full-stack developer

Platform, API, user interface

Data engineer

Dataset pipeline and quality control

Our advantage is not the technology — it is the data

Anyone can write a CNN. Nobody else in the country holds 5,000+ expert-described analyses of Uzbek rocks, 20+ completed research projects, 12 exploration projects and 27 commercial contracts.

THE ASK

$100,000 investment, the acceleration programme, and a route to industrial customers

35%	Dataset creation and expert annotation
30%	Engineering team
20%	Compute infrastructure (GPU)
10%	Microscope imaging equipment
5%	Intellectual property and legal setup

What we have

- Expert-described reference material
- A working demo prototype
- Industrial partners and a test site

What we need

- Engineering capacity
- Product-market mentorship
- A path to international markets

Live demo: yuldoshevz1995-spec.github.io/Petrolog-AI